

Fig. 2: Circuit of whisker sensor



Fig. 3: Whisker for robot

shown in Fig. 2. Pull-up resistor R4 is used because the LM339 has only

open-collector outputs. The working is simple. Initially, when there is no interruption, the output of the comparator remains low. As and when

any obstacle disturbs the guitar string of the whisker switch, pin 4 of IC1 goes low and output pin 2 of the first comparator of IC1 becomes high momentarily. We have described only one comparator but you can use the other three comparators similarly. The brain (microcontroller) of the robot can now read the status of the whisker switch through this comparator output.

Assemble the circuit on a small general-purpose PCB and connect all the four whiskers to your microcontroller-based robot as shown in Fig. 3. ●

TRANSISTORISED LOGIC PROBE FOR TTL

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A logic probe is a handheld pen-like probe used for analysing and troubleshooting the logical states '0' or '1' of a digital circuit. While most logic probes are powered by the circuit under test, some probes use batteries. These can be used for either TTL (transistor-transistor logic) or CMOS (complementary metallic oxide semiconductor) integrated circuit devices. The circuit described here can be used for TTL logic only, and shows the presence of either state (logic 1 or 0).

As shown in Fig. 1, the logic probe circuit is built around four transistors and a few passive components. It uses two LEDs that show logic states 1 (green) and 0 (red). The green LED glows when the voltage at probe tip exceeds about 2.4V and the red LED glows when this voltage falls below about 1.2V, which is adequate to detect the normal TTL levels. When no LED glows, it means the probe is in suspended state, i.e., it is neither showing logic 1, nor logic 0.

Working of the logic probe circuit is simple and can be divided into three states: The probe is suspended, the probe's tip is at logic 1 and the probe's tip is at logic 0.

When the probe is in suspended condition, the junction voltage of resistors R1 and R2 is almost equal to that obtained when R1 and R2 alone form

a voltage divider bias for the supply voltage. Under this condition, the base-emitter junction of transistor T1 is forward biased, so the voltage drop across resistor R5 (>0.6V) is enough to drive transistor T3 into conduction, which, in turn, cuts off transistor T4. As a result, the red LED (LED2) does not glow. The current flowing through R3 is nearly equal to that through R5. However, this does not produce enough voltage drop (0.3V-0.6V) across resistor R3 to make transistor T2 conduct. As a result, the green LED (LED1) does not glow. Thus neither of the LEDs glows when the probe is in suspended condition.

When the probe tip is at logic 1, transistor T3 forward biases to cut-off transistor T4. But, this voltage should exceed 2.4V (normally TTL logic 1 is well above this voltage), this will cause a greater current to flow through emitter resistor R5 of T1, which also flows through collector resistor R3 causing a larger voltage drop across R3. Thus transistor T2 conducts and the green LED (LED1) glows to indicate the presence of logic 1 (high) at the probe tip.

When the probe tip is at logic 0, transistor T3 cuts off to make transistor

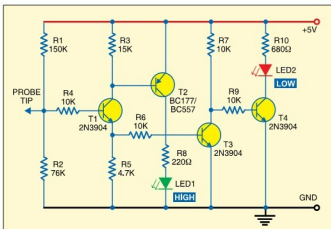


Fig. 1: Circuit of transistorised logic probe

